

CHOOSING THE BEST **WI-FI ROUTER** FOR YOUR HOME

Paromik Chakraborty

In our daily lives, one thing we do not want to compromise with is our Internet connection. Most Internet service providers provide a router along with their service. However, to get the best out of your network, it is better to handpick your own router. In this article we explore the various factors that must be considered while picking the right Wi-Fi router for homes.

Important router features

The choice of a Wi-Fi router depends on various factors of use: area to be covered, number of users and devices to be connected to the network, Internet connection speed, usage mode (moderate or data-intensive) and so on. For homes, speed and coverage range are the primary consideration factors. Having good bandwidth support and a secure connection also play important roles in the router's performance. Given below is what you should look for while buying a router.

Dual-band support. Routers mainly function on two bandwidth frequencies: narrower 2.4Ghz and much wider 5.0Ghz. The more common single-band Wi-Fi routers tap into 2.4Ghz bandwidth only to transmit data. The challenge

with this is that it is also tapped in by other electronic or wireless products such as phones and tablets, cluttering the channel. As a result, while smaller data packets can still pass through, heavier data (like videos or large files) fail to transmit properly, deteriorating Wi-Fi performance.

A dual-band router, on the other hand, taps into both 2.4Ghz and 5.0Ghz bands. Small data packets are passed through 2.4Ghz bandwidth, while larger data travels through 5.0Ghz channel, which has much less traffic. Therefore users experience better media streaming or online gaming using a dual-band device. However, a dual-band router is costlier.

Latest standard and speed. Wi-Fi works on IEEE 802 group of standards. IEEE 802.11ac is the most upgraded standard at the moment—compared to older standards (802.11n, b, g or a), it is faster and more optimised for network operations. However, routers supporting 802.11ac are costlier. If the intended usage is not too data-intensive, then routers supporting 802.11n will suffice.

Wi-Fi's hardware and build quality also play an inevitable role in speed capacities. Check the highest speed mentioned in the specifications of the product—higher the speed, higher will be the price. A 300mbps capacity is a decent option for all kinds of Internet and networking operations, including live streaming of videos, moderate gaming and Web browsing. To stream 4K videos or high-end online games, it is advisable to go for a 600mbps or higher device.

Antennae and coverage. A higher

number of antennae means more channels for data transmission and reception, ensuring better data flow rates. Single-band routers have only 2.4Ghz bandwidth-compatible antennae.

On the other hand, dual-band routers may come with dual-band-compatible antennae (capable of data transfer across both 2.4Ghz and 5.0Ghz bands), or have separate antennae, each assigned to individual bandwidths.

For a small, single-floor premise (within 55.7sqm), a single-antenna dual-band router or a two-antennae single-band router will suffice.

For a larger area, for both categories of devices, a higher number of antennae (two to three) will have advantages in terms of signal strength, coverage and speed.

Antennae strength is measured in dBi unit, which essentially indicates signal strength. For a medium-size area, 5dBi is enough. If the premise has more walls and obstructions, or is larger, go for 9dBi specification.

Security. For a home setup, Wi-Fi Protected Access 2 (WPA2) with at least 128-bit Advanced Encryption Standard and Pre-Shared Key (PSK) encryption compatibility is compulsory. Good practices like using a strong Wi-Fi password, upgraded internal firewalls and updated firmware also play crucial roles in network security.

More features. Choosing a router with more Ethernet Ports (four or more) can be advantageous. This allows you to connect more devices parallelly and provide backup if the main port malfunctions.

Some routers come with a USB